



National Institute for Public Health
and the Environment
Ministry of Health, Welfare and Sport

Noisemodelling in the Netherlands

Rob van Loon

RIVM



RIVM

- › Research institute
- › Health and the environment
- › Noise:
 - Centre for environmental quality
 - Data, software, methods: numbers
 - Centre for Sustainability, environment, and health
 - Health impact assessment
 - Dose/impact relations





Noise projects

- › Noise monitor (10+ years)
- › Train Vibrations
- › Airplane noise
- › Wind turbines
- › Noise measuring instructions
- › Noise calculation instructions
- › Noise calculation software

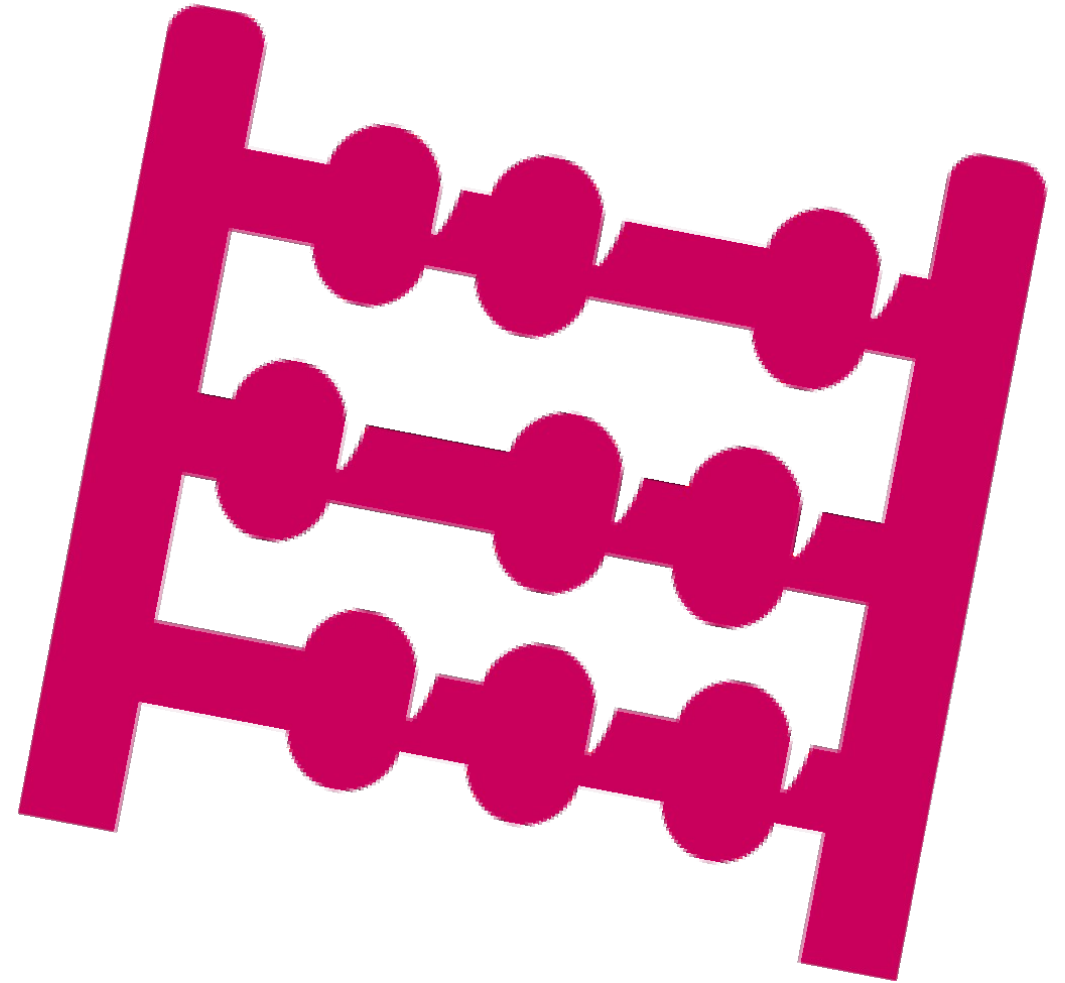




Software

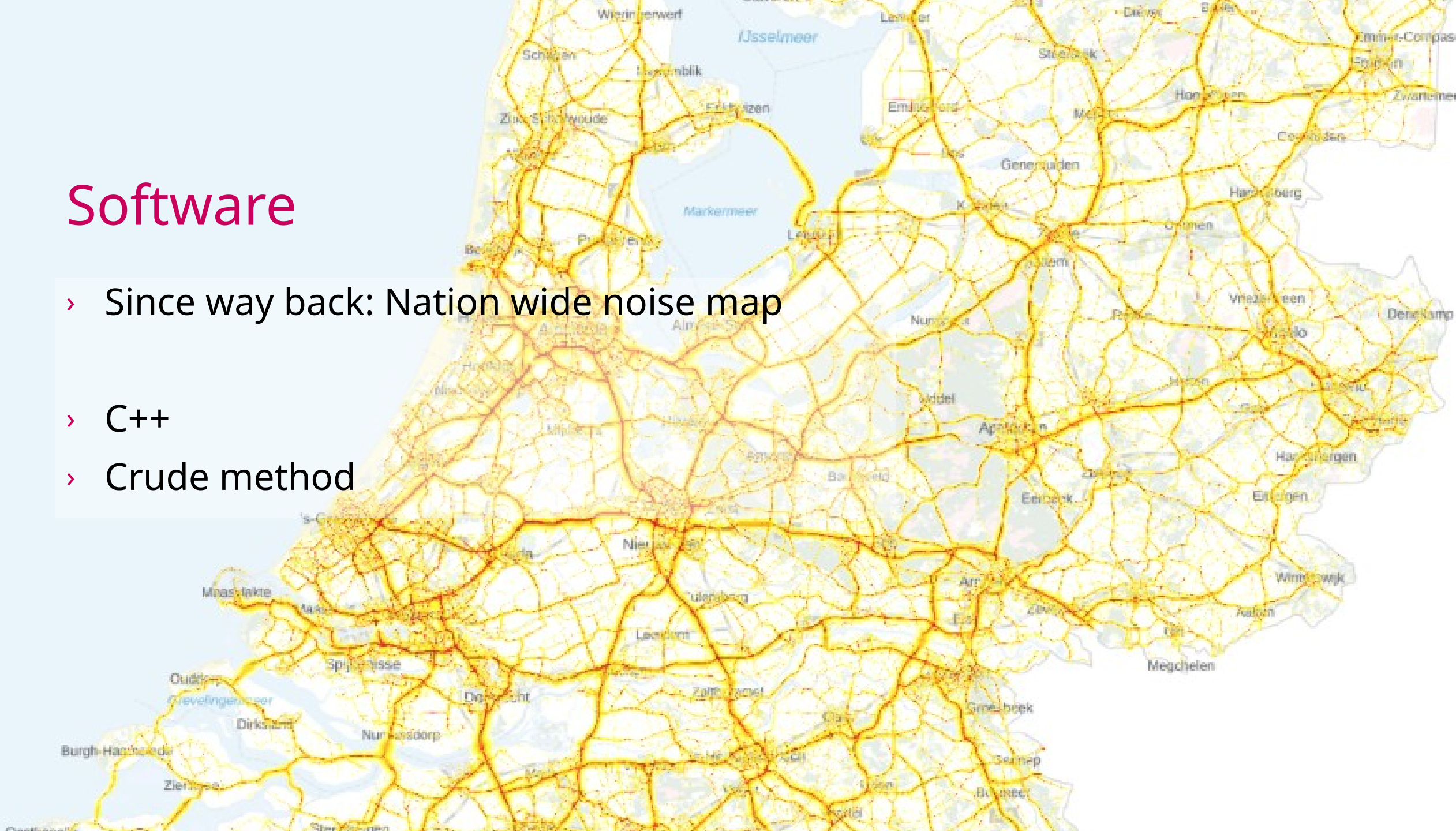
- › Since 2024: Noise Calculation Software: SRM2 (ISO 9613)
 - Road
 - Railway
 - Industry

(not yet publiced as open source project)



Software

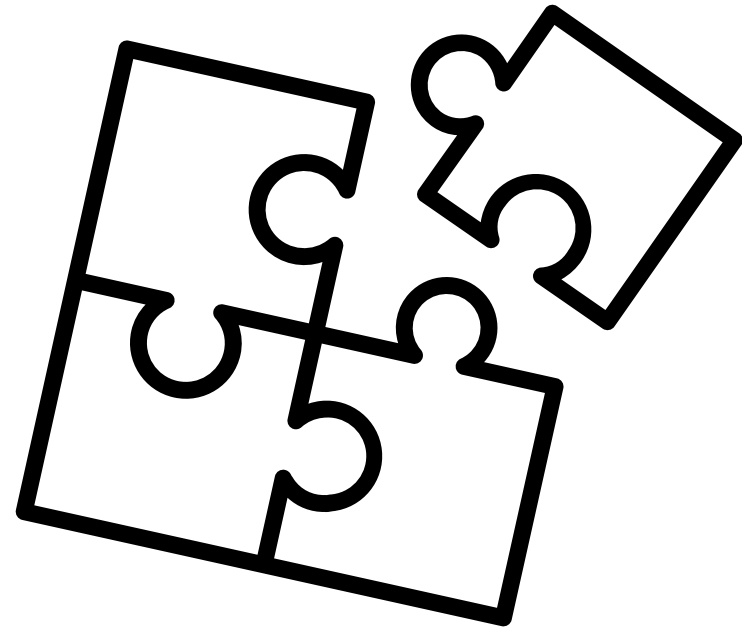
- › Since way back: Nation wide noise map
- › C++
- › Crude method





Software

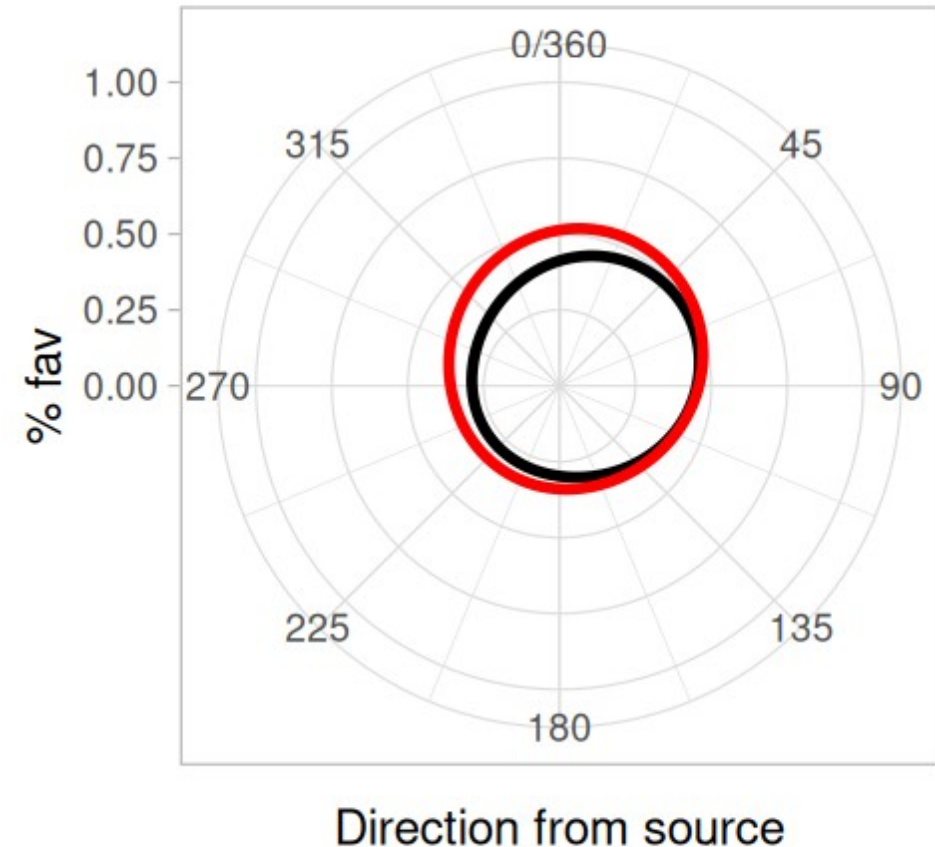
Missing: CNOSSOS method





CNOSSOS in the Netherlands

- › Mostly CNOSSOS like the rest of Europe
- › NoiseModelling would be a great fit. (Passing ISO-tests? Great!)
- › Dutch cars, Dutch pavements, Dutch trains
- › Dutch weather



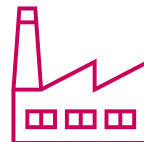


Noisemodelling in the Netherlands

- › Implement Dutch emission parameters.
 - Modify JSON parameter files.
- › Implement ratio between homogenous and favorable conditions.
 - Different for day and evening/night.
 - As a *continuous* function of compass direction.

$$p = \begin{cases} 0,34 - 0,1 \sin(\xi + 35) + 0,045 \sin^2(\xi + 35) & \text{voor de dagperiode} \\ 0,40 - 0,1 \sin(\xi + 60) + 0,035 \sin^2(\xi + 60) & \text{voor de avond en nachtperiode} \end{cases}$$

- › Not for research, for production work!





Code contribution

Crossos NL #1039

Merged [pierromond](#) merged 48 commits into [Universite-Gustave-Eiffel:main](#) from [loonvr:crossos_NL](#) 2 days ago

Conversation 21

Commits 48

Checks 2

Files changed 31

+8,275 -3,373

loonvr commented on Aug 21

Contributor

...

With this PR we welcome NoiseModelling into the Netherlands.
I have added Dutch emission parameters for roads and rail traffic.

- For rail, the user can specify the Dutch train reference (eg. `NL_CAT12`) and track parameters (eg. `NL_BB1`). I've also

Reviewers

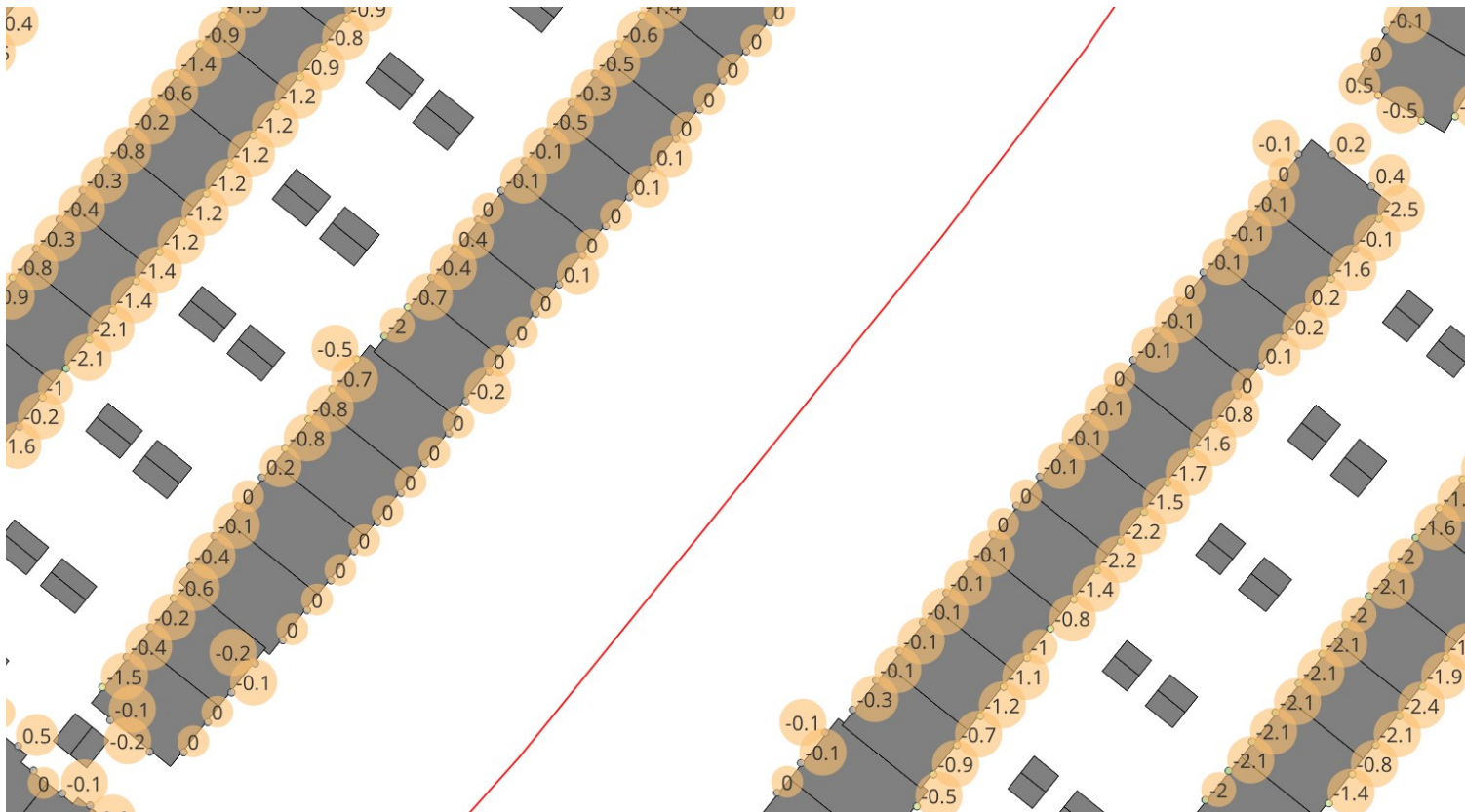
[nicolas-f](#)

[pierromond](#)

Assignees



Comparing



City and Country side

First line buildings: very good match

Afterwards: small differences add up

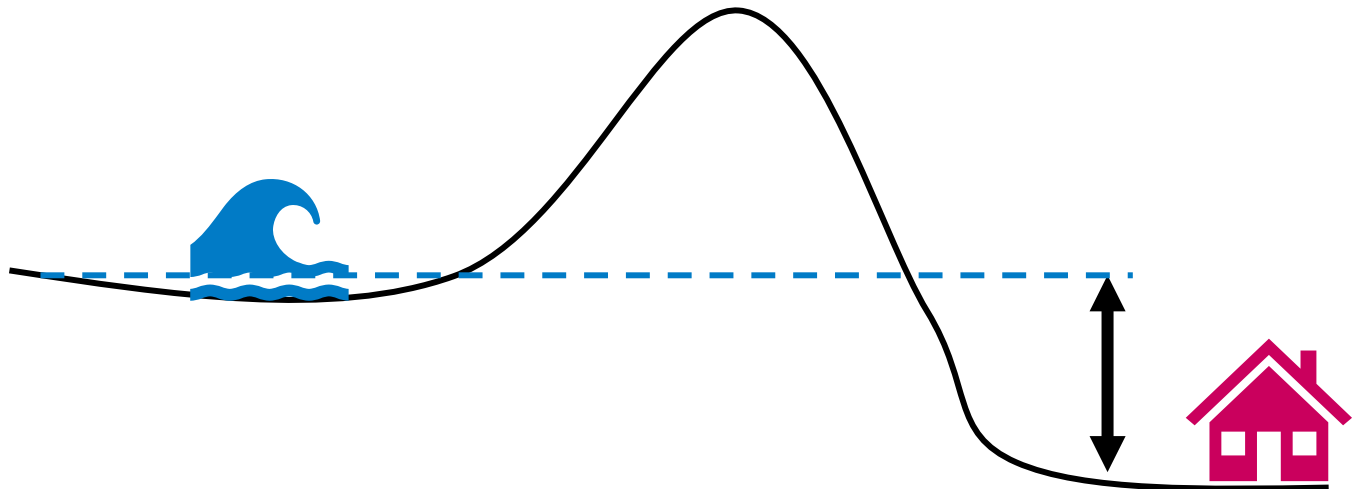
Height: lines versus points

Pathfinder algorithm



Height: important

- › Acts like extra barrier
- › Ground effect (absorption/reflection)
- › Absolute height



Front end

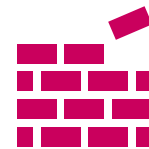
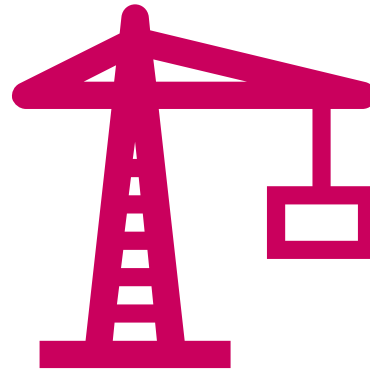
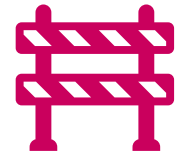
- › Focus on calculation software, not GUI or front end
- › dBvision: QGIS plugin (extra module on existing plugin)
- › ...





Model building

- › Buildings
- › Ground
- › Height
- › Road + rail
- › Screens and barriers



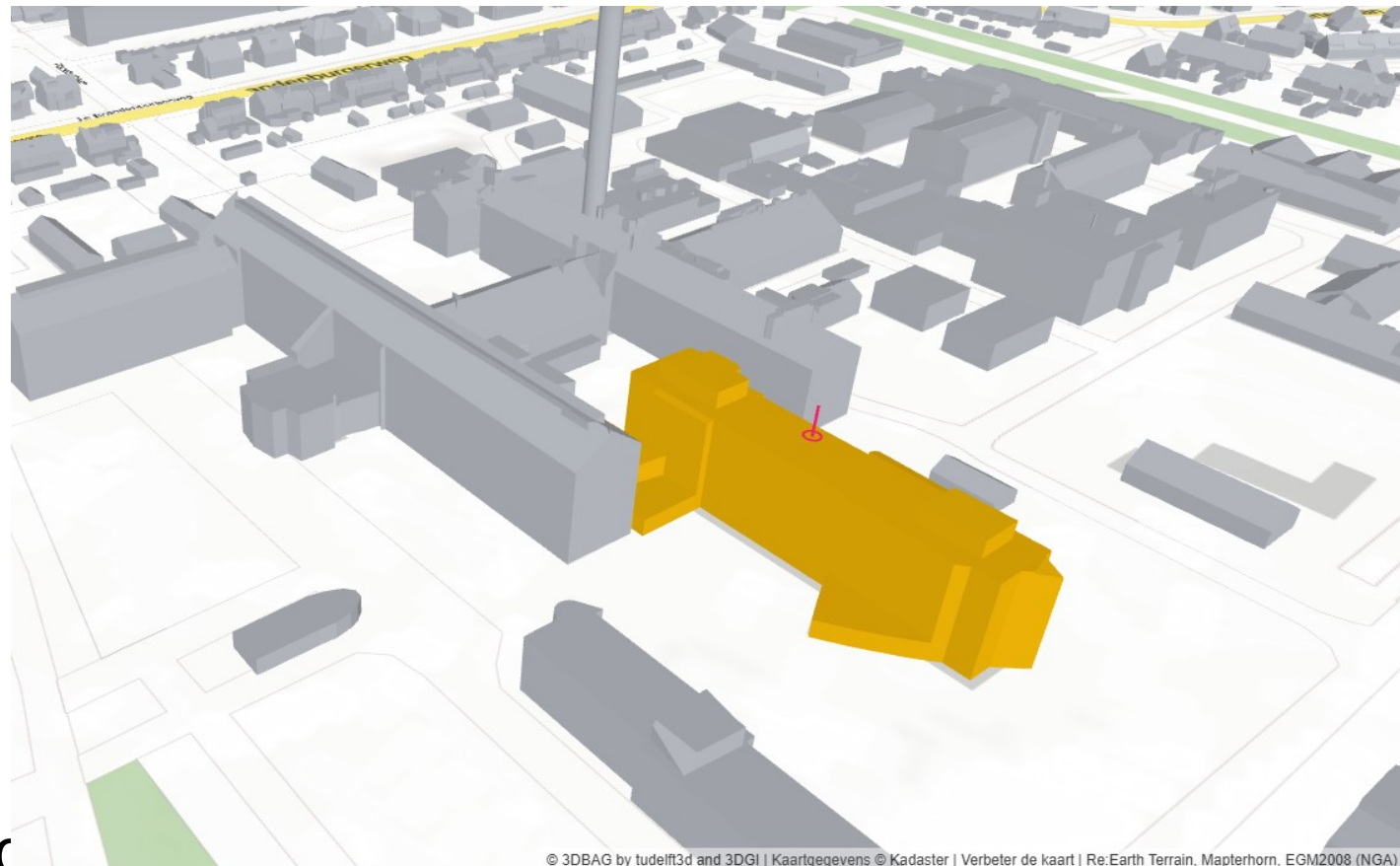


Open data

- › 3D model Netherlands
- › Road intensity highways

Not open, but available:

- › Rail intensity
- › Provincial and city roads
- › Screens and barriers
- › Pavement and maximum speed



© 3DBAG by tudelft3d and 3DGI | Kaartgegevens © Kadaster | Verbeter de kaart | Re:Earth Terrain, Maptherhorn, EGM2008 (NGA)

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- 343.000 receivers
- 90.000 buildings
- 57.000 ground areas
- 200 km road
- $\sim 15 \times 19 \text{ km} = 285 \text{ km}^2$
- 7 year old Intel i5
- 2 hours calculating
- Tweaked down settings



What is next?

- › Testing and comparing
- › Tell the world
- › Upcoming END – open source and free alternative
- › Noise Calculation in the Netherlands
- › Contribute to NoiseModelling